

INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD
DEPARTMENT OF MATHEMATICS

PROBLEM SHEET 3

MA 2130 : Complex Variables

Year: 2026

1. (a) Prove that, if $f(z)$ is differentiable at z_0 , then

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + h(z)(z - z_0)$$

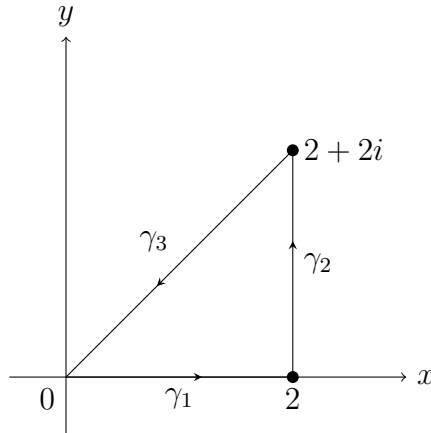
where $\lim_{z \rightarrow z_0} h(z) = 0$.

- (b) If $f(z)$ is differentiable at z_0 , then show that $f(z)$ is continuous at z_0 .
2. (a) If $f(z)$ is analytic from a domain $D \rightarrow \mathbb{R}$, then show that $f(z)$ is a constant function on D .
- (b) If $f(z)$ and $\overline{f(z)}$ are analytic on a domain D , then show that $f(z)$ is a constant function on D .
3. If $f(z) = u(x, y) + iv(x, y)$ is an analytic function on a domain D , then show $u(x, y)$ and $v(x, y)$ are harmonic functions. Is the converse true?
4. Suppose $f(z) = e^x(x \cos y - y \sin y) + iv(x, y)$ is an analytic function. Determine a possible choice for the function $v(x, y)$.
5. If n is real and $f = r^n(\cos(n\theta) + i \sin(n\theta))$, then show f is analytic everywhere except possibly at $r = 0$.
6. Let $z = x + iy$. Then, which of the following functions are analytic on $\mathbb{C}$?
- (a) $f(z) = x - iy$
- (b) $f(z) = e^x(\cos y - i \sin y)$
- (c) $f(z) = e^{-x}(\cos y - i \sin y)$
- (d) $f(z) = xy + iy$
7. Determine the real-valued functions u and v such that $z^3 = u(x, y) + iv(x, y)$, where $z = x + iy$. Verify that both u and v satisfy Laplace's equation.
8. Suppose that $u_k(x, y) = x^3 - kxy^2 + 12xy - 12x$ for some $k \in \mathbb{C}$. Determine all values of k such that u_k is the real part of an analytic function f_k on $\mathbb{C}$.
9. Show that the only holomorphic function $f : \mathbb{C} \rightarrow \mathbb{C}$ which is of the form $f(x + iy) = u(x) + iv(y)$ is given by $f(z) = \lambda z + a$ for some $\lambda \in \mathbb{R}, a \in \mathbb{C}$.
10. Suppose that $f(z) = u(x, y) + iv(x, y)$ is a holomorphic function such that

$$2u(x, y) + v(x, y) = 5 \text{ for all } z = x + iy \in \mathbb{C}.$$

Show that f is a constant function.

11. Show that if $f : \mathbb{C} \rightarrow \mathbb{C}$ is holomorphic function such that $\operatorname{Re}(f(z))$ is constant, then f is a constant function.
12. Show that $f(z) = z^4 = r^4 e^{4i\theta}$ satisfies the polar Cauchy-Riemann equations at a point $z = re^{i\theta}$ with $r \neq 0$. Write down $f'(z)$ using the polar derivative formula.
13. Compute $\int_{\Gamma} \bar{z}^2 dz$, where $\Gamma = \gamma_1 \cup \gamma_2 \cup \gamma_3$.

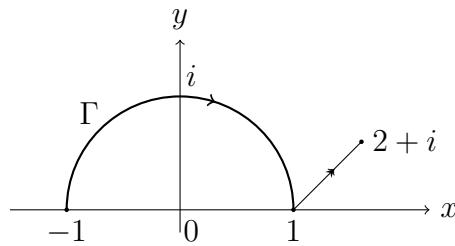


14. Find an upper bound for

$$\left| \int_C \frac{e^z}{z^2 + 1} dz \right|,$$

where C is the circle $|z| = 2$ traversed once in the counterclockwise direction.

15. Compute the integral $\int_{\Gamma} \cos z dz$, where Γ is the contour shown in the figure below.



$$\Gamma = \gamma_1 \cup \gamma_2,$$

where

$$\gamma_1(t) = e^{i(\pi-t)}, \quad 0 \leq t \leq \pi,$$

and

$$\gamma_2(t) = 1 + t(1+i), \quad 0 \leq t \leq 1.$$

Hint: Use the notion of anti-derivative.